

CREATE Module:
Perturbation Theory
Lecture 3: Order symbols and
asymptotic
Marek Stastna

Some useful review: Local Approximations

Marek Stastna

Outline

- ✿ What are local approximations?
- ✿ Taylor series versus Taylor polynomials: convergence and error.
- ✿ Order notation.
- ✿ Uses of local approximations.
- ✿ Computer algebra (e.g. Maple) is largely extra but could be a useful part of your module work.

Maple

- ✿ Maple is a symbolic algebra package with a decent graphical interface.
- ✿ While clumsy in some ways, Maple takes away the “I missed a factor of two in step 8 of the calculation” aspect of approximate analysis.
- ✿ It is thus a powerful time saver, as well as a means to tackle new problems (which all grad students want to do).
- ✿ There are no "style points" for doing something by hand!

Local Approximations

- ✿ A local approximation uses information at one or more points to derive an approximation of a function over an interval.
- ✿ For example the classical linear approximation uses the value of a function and the value of its derivative at a point to construct a “line of best fit”.

Taylor Series and Polynomials

$$P_{n,a}(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a)(x-a)^k$$

Taylor polynomial

- What do we mean by “good approximation”?

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(a)(x-a)^k$$

Taylor series

- What do we mean by the equal sign (i.e. mathematical equivalence)?

Taylor Polynomials

$$P_{n,a}(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a)(x-a)^k$$

- ✦ Taylor polynomials are useful because we can control the error they make in replacing the original function by the polynomials.
- ✦ This is the content of Taylor's Theorem (most first year calculus books will at least state it).

Taylor Polynomials: Error

$$P_{n,a}(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a)(x-a)^k$$

$$R_{n,a} = \frac{1}{(n+1)!} f^{(n+1)}(t)(x-a)^{n+1} \text{ where } a < t < x$$

- ✿ This is an explicit formula for the error, though with a twist, since “t” is not specified.
- ✿ In practice, an upper bound on error is all we need, though even this cannot be guaranteed in “closed form”.

Taylor Polynomials: Error

$$P_{n,a}(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a)(x-a)^k$$

$$R_{n,a} = \frac{1}{(n+1)!} f^{(n+1)}(t)(x-a)^{n+1} \text{ where } a < t < x$$

- ✿ Mathematics does tell us more. If the original function is smooth enough so that the $(n+1)$ th derivative is continuous on the interval $[a, x]$ then it is guaranteed to have a maximum. The error thus has the form

$$R_{n,a} = M(x-a)^{n+1}$$

Taylor Polynomials: Error

$$P_{n,a}(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a)(x-a)^k \quad R_{n,a} = M(x-a)^{n+1}$$

We will define this in two slides

- ✿ We would thus say that the error made is “order $n+1$ ”.
- ✿ This rather amazing fact is so basic to most of what we do post first year calculus that we almost never think about it.
- ✿ It says that calculating one more term of a Taylor polynomial can make an approximation much more accurate!

Taylor Polynomials: Error

$$P_{n,a}(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a)(x-a)^k \quad R_{n,a} = M(x-a)^{n+1}$$

- ✿ There are two caveats:
- ✿ The first is whether we can actually calculate the next derivative, e.g. $\sin(\sqrt{\tan x^2 + \exp(-\sin x)})$
- ✿ The second is the constant in front of the error (i.e. is M tiny or huge), since in practice we almost always want to approximate at a finite distance from $x=a$.

- We begin with defining the big Oh and little oh symbols:

$$1. f(x) = \mathbf{O}(g(x)) \text{ as } x \rightarrow a \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \text{constant}$$

$$2. f(x) = \mathbf{o}(g(x)) \text{ as } x \rightarrow a \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0$$

- The two symbols obviously denote a comparison, with the little oh symbol being a stronger statement.
- In fact the big Oh is a really weak statement since $10^9 = O(1)$, a statement that seems on first sight kind of crazy.
- The Taylor polynomial can be written two ways:

$$\bullet P_{n,a}(x) = \sum_{n=0}^N \frac{f^{(n)}(a)}{n!} (x-a)^n + O((x-a)^{n+1}) \text{ or the less common}$$

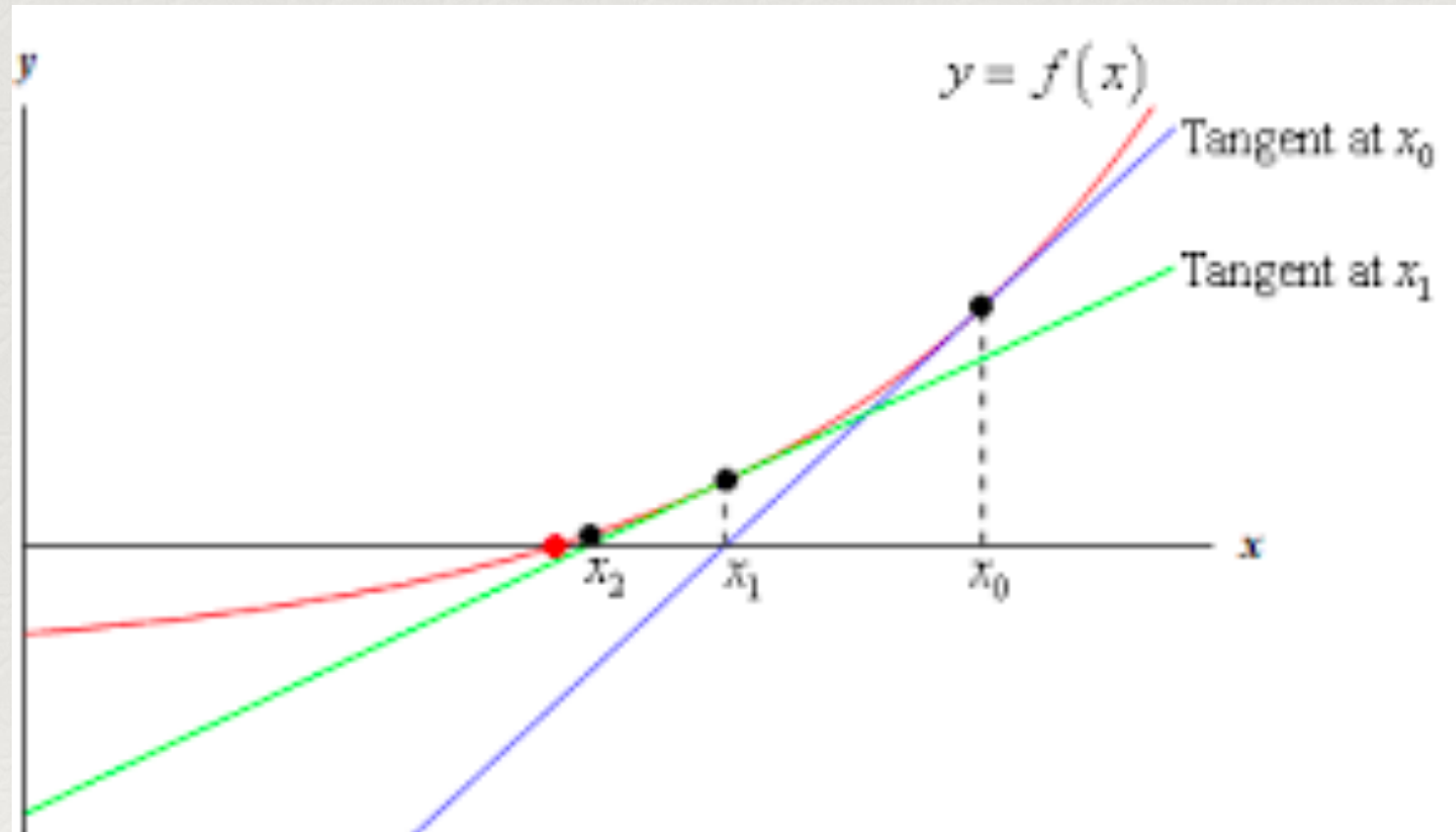
$$\bullet P_{n,a}(x) = \sum_{n=0}^N \frac{f^{(n)}(a)}{n!} (x-a)^n + o((x-a)^n)$$

Taylor series uses

- ✿ But why do we want to compute approximations anyway?
- ✿ Well it could be to evaluate a transcendental function like the exponential or sine and cosine.
- ✿ Or it could be to solve algebraic equations, and it is this example which we look at next using Newton's Method.

Newton's method concept

- ✿ Newton's method repeatedly replaces $f(x)$ by its linearization in an effort to solve $f(x)=0$.



Newton's Method

- ✦ Newton's method is a great example of applied mathematics:
https://en.wikipedia.org/wiki/Newton%27s_method
- ✦ The first reason is that when it works it works in an amazing way.
- ✦ The second reason is that when it fails, it does so in an equally spectacular way.
- ✦ Newton's method uses the linear approximation repeatedly to get a better and better approximation (when it converges).

Try computing examples that work and don't work

Finite Differences

- ✿ Real world data, say we measure something every second, is discrete.
- ✿ This means that computing the Taylor polynomial based on measurements is impossible since we can't compute the derivative.
- ✿ However a Taylor polynomial can still be used to compute approximations in an indirect way.
- ✿ This is an example of how an every day mathematical object can be used in an abstract way.

Finite Differences

- ✿ Say we know the values of a function at three points.
- ✿ Though we don't know the derivatives the Taylor polynomial can still be formally written down.

$$f(x_0 \pm \Delta x) = f(x_0) \pm f'(x_0)\Delta x + \frac{1}{2}f''(x_0)\Delta x^2$$

- ✿ Note the symmetry in the expression when the three points are regularly spaced.
- ✿ The polynomial approximation involves three unknowns.

Finite Differences

- ✦ This is three linear equations for three unknowns, (a,b,c), so easy.

$$p(x) = a(x - x_0)^2 + b(x - x_0) + c$$

$$p(x_0 - \Delta x) = f(x_0 - \Delta x)$$

$$p(x_0) = f(x_0)$$

$$p(x_0 + \Delta x) = f(x_0 + \Delta x)$$

- ✦ Can also estimate the derivative at x_0

Try computing a one sided estimate of f'

$$f'(x_0) = \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

Local Approximations

- ✿ In summary local approximations, and Taylor polynomials in particular, can be used directly to get numerical values.
- ✿ They can also be used indirectly to describe data in a way that is consistent with the theoretically better posed theory of Calculus on the real line.
- ✿ We are still left with the notion of convergence.

Taylor Series

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(a)(x-a)^k$$

- ✿ The notion of convergence of a power series is a powerful mathematical idea (it isn't often presented that way).
- ✿ It states that the object on the left (a function) is the same, for mathematical purposes, as the object on the right.
- ✿ The series is thus a “representation” of the function, and if the function is, for example, a solution to a DE, this could be a useful tool.

Taylor Series

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(a)(x-a)^k$$

- ✿ Convergence can be pointwise (a weak criterion) and uniform (a strong criterion).
- ✿ Uniform convergence allows for differentiation of the series term-wise (integration term wise is almost always OK).
- ✿ The radius of convergence theorem (for $a=0$) says that if a series converges at a point b , then it converges uniformly on $[-c, c]$ where $c < b$. A very powerful result.

Padé Approximants

This is an example of “beyond” topic
for the lab book

- ✿ For certain types of functions like periodic functions and even oscillations with a slowly varying envelope Taylor series do a bad job.
- ✿ However it is easy to generalize the Taylor series notion to give a ratio of polynomials.
- ✿ Such approximations are called Padé approximants.

$$f(x) \approx R(x) = \frac{\sum_{j=0}^m a_j x^j}{1 + \sum_{k=1}^n b_k x^k}$$

Computing Padé Approximants

- Formally we would say:

$$f(0) = R(0)$$

$$f'(0) = R'(0)$$

...

$$f^{(m+n)}(0) = R^{(m+n)}(0)$$

- In practice polynomial division is used, and computer algebra packages like Maple excel at this, we will see more in the lab.

- We use Taylor polynomials when we solve ODEs (and PDEs) numerically.
- We also make use of a different approximation concept when we do perturbation theory.
- Return to the dimensionless damped oscillator:

$$\frac{d^2y}{ds^2} + a\frac{dy}{ds} + y = 0$$
where $a = \gamma T$ and we assume $a \ll 1$
- The solution can be written formally as $y(s; a)$.
- The semi-colon denotes that s is an independent variable and a is a parameter (think of it as something measured).
- The approximation we would like now is to find $y(s)$ as $a \rightarrow 0^+$.
- This is what we did without thinking too deeply about it when we did RPT. Let's take a half lecture to explore the math behind it.

- If we are going to consider small perturbations we should come up with sensible ways to compare terms.
- We want something more general than $<$ and $>$.
- We begin with defining the big Oh and little oh symbols (which you may have seen in Calculus):

$$1. f(x) = \mathbf{O}(g(x)) \text{ as } x \rightarrow a \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \text{constant}$$

$$2. f(x) = \mathbf{o}(g(x)) \text{ as } x \rightarrow a \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0$$

- The two symbols obviously denote a comparison, with the little oh symbol being a stronger statement.
- In fact the big Oh is a really weak statement since $10^9 = O(1)$, a statement that seems on first sight kind of crazy.

$$\mathbf{1.} \ f(x) = \mathbf{O}(g(x)) \text{ as } x \rightarrow a \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \textit{constant}$$

$$\mathbf{2.} \ f(x) = \mathbf{o}(g(x)) \text{ as } x \rightarrow a \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0$$

- For well behaved functions with a typical finite value of “a” there isn’t much to say about these symbols.
- In particular for any two continuous functions on a closed interval [p,q] condition 1 is met at all points in (p,q).
- So why bother? Well consider the pair of functions: $f(x) = x$, $g(x) = x^2$ in the limit $x \rightarrow \infty$
- We know $g(x)$ grows faster than $f(x)$ so that: $\lim_{x \rightarrow \infty} f(x)/g(x) = 0$ so $f(x) = o(g(x))$
- Now consider the limit as $x \rightarrow 0$, now $\lim_{x \rightarrow 0} g(x)/f(x) = 0$ as $x \rightarrow 0$ so $g(x) = o(f(x))$
- This second fact reflects to notion that squaring a small number makes it smaller, meaning that $g(x)$ goes to zero faster as x tends to zero.
- There are more “tricky” properties in books on asymptotic.
- Now let’s try to use these order symbols for something.

- When we looked at RPT for polynomials we wrote things like: $(x_0 + \epsilon x_1 + \epsilon^2 x_2 + \dots)^2$ using the definition of the big Oh and little Oh we can now make this more precise.

$$x = x_0 + \mathbf{O}(\epsilon) = x_0 + \epsilon x_1 + \mathbf{O}(\epsilon^2) = x_0 + \epsilon x_1 + \epsilon^2 x_2 + \mathbf{O}(\epsilon^3)$$

$$x = x_0 + \mathbf{o}(1) = x_0 + \epsilon x_1 + \mathbf{o}(\epsilon) = x_0 + \epsilon x_1 + \epsilon^2 x_2 + \mathbf{o}(\epsilon^2)$$

- Both hold as $\epsilon \rightarrow 0$, with the first expression being more common.
- Basically the order symbol tells you what the coefficient of the next term will be.
- A special version of big Oh is the statement of asymptotic equivalence:

$$f(x) \sim g(x) \text{ as } x \rightarrow a \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 1$$

- The most famous example of asymptotic equivalence that I know of is: $\sin(x) \sim x$ as $x \rightarrow 0$.

$$f(x) \sim g(x) \text{ as } x \rightarrow a \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 1$$

- If we have $f(x) \sim g(x)$ as $x \rightarrow a$ then $f(x) - g(x) = o(g(x))$ as $x \rightarrow a$
- And this is where things get a bit subtle.
- Say $f(x) = \exp(x) + x$, $g(x) = \exp(x)$, $f(x) - g(x) = x$ and consider the limit $x \rightarrow \infty$
- We can clearly see that the difference is little oh of $g(x)$, BUT the difference does not go to zero.
- In fact the difference grows without bound!
- What this is really saying is that when we compare we should always say what we are comparing with respect to.
- $f(x) - g(x)$ is large when x is large when compared to 1, but it is vanishingly small when compared to $\exp(x)$.

Definition 4.1.4 (Asymptotic Series). To say that

$$g(x) \sim x^4 - 3x^2 - 2x + \cdots \text{ as } x \rightarrow \infty,$$

means the following:

1. $g \sim x^4$, i.e. $\frac{g}{x^4} \rightarrow 1$ as $x \rightarrow \infty$,
2. $g - x^4 \sim -3x^2$, i.e. $\frac{g-x^4}{-3x^2} \rightarrow 1$ as $x \rightarrow \infty$,
3. $g - x^4 + 3x^2 \sim -2x$, i.e. $\frac{g-x^4+3x^2}{-2x} \rightarrow 1$ as $x \rightarrow \infty$,

etc. The series on the right hand side is an example of an asymptotic series. In the series the fastest growing term comes first. Each successive term must grow more slowly than the preceding term.

Asymptotic series are very useful for finding approximate values of integrals and functions, which we consider next.

- The above is taken straight from a set of graduate notes, and indeed something like it can be found in many books on asymptotic expansions and perturbation theory.
- It does make an assumption about the reader's point of view that I don't think is a good one for today's student.
- So I am going to go over exactly what these integrals we may want expansions of are, assuming no background.

- The Gaussian $f(x) = \exp(-x^2)$ is an infinitely differentiable (i.e. well behaved), bounded function that crops up in many mathematical and applications contexts.
- It is well known that this function does not have an elementary integral, but returning to the geometric definition of the integral we can imagine that there is some utility in being able to calculate:

$$A(x) = \int_{-\infty}^x f(m) dm$$

- Now consider a class of Gaussians $f(x) = \exp(- (x/\epsilon)^2)$.
- One can imagine wishing to know the special behavior of $A(x; \epsilon)$ in certain special limits like $\epsilon \rightarrow 0, \epsilon \rightarrow \infty$ (really sharp or really wide).
- In fact before numerical solutions were readily available many DEs were solved “in quadrature”, or in terms of integrals and special limit of these integrals were sought in lieu of solutions, and this brings up the following:

4.2.1 The Exponential Integral

The exponential integral function $\text{Ei}(x)$ is defined by:

$$\text{Ei}(x) = \int_x^\infty \frac{e^{-t}}{t} dt \quad \text{for } x > 0. \quad (4.2.1)$$

This is not very useful as it stands – can we find a useful approximation? Successively integrating by parts gives

$$\text{Ei}(x) = e^{-x} \underbrace{\left(\frac{1}{x} - \frac{1}{x^2} + \frac{2!}{x^3} - \frac{3!}{x^4} + \cdots + \frac{(-1)^{n-1}(n-1)!}{x^n} \right)}_{S_n(x)} + \underbrace{(-1)^n n! \int_x^\infty \frac{e^{-t}}{x^{n+1}} dt}_{R_n(x)}. \quad (4.2.2)$$

As $n \rightarrow \infty$, $S_n(x)$ gives a divergent series as is easily seen from the ratio test. The ratio of the $(m+1)^{\text{st}}$ and m^{th} terms is

$$\frac{\frac{(-1)^m m!}{x^{m+1}}}{\frac{(-1)^{m-1} (m-1)!}{x^m}} = \frac{m}{x} \rightarrow \infty \text{ as } m \rightarrow \infty \quad (4.2.3)$$

- So as a traditional series approximation this seems useless.
- But for a fixed m the asymptotic series is a good approximation as x gets large (details on next slide).
- Historically this is what many mathematical physics problems were reduced to; special limits (i.e. we know how $\text{Ei}(x)$ behaves for large x).

Have

$$\begin{aligned} |\operatorname{Ei}(x) - S_n(x)| &= |R_n(x)| \\ &\leq n! \int_x^\infty \frac{e^{-t}}{t^{n+1}} dt \\ &\leq \frac{n!}{x^{n+1}} \int_x^\infty e^{-t} dt \end{aligned}$$

so

$$|\operatorname{Ei}(x) - S_n(x)| \leq \frac{n!}{x^{n+1}} e^{-x} \rightarrow 0 \text{ as } x \rightarrow \infty. \quad (4.2.4)$$

- It is good to be aware that results like this are available as they could be used to test numerical methods in some cases.
- But they probably have to be balanced by actual use of numerics in practice.